

A PROJECT TITLE
ON
**EARLY-STAGE DIABETES RISK PREDICTION USING RANDOM FOREST AND XGBOOST
CLASSIFIERS**

Project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF ENGINEERING

Name of Title: **EARLY-STAGE DIABETES RISK PREDICTION USING RANDOM FOREST AND XGBOOST CLASSIFIERS**

1. Abstract:

Diabetes is one of the most common chronic health conditions worldwide and can cause serious complications when it is not detected and managed at an early stage. Early identification of diabetes risk can help individuals seek timely medical advice and adopt preventive measures. This project, **Early-Stage Diabetes Risk Prediction Using Random Forest and XGBoost Classifiers**, focuses on developing a Machine Learning-based system for predicting the likelihood of diabetes using patient information and common symptoms. The system considers features such as age, gender, frequent urination, excessive thirst, sudden weight loss, weakness, family history, and other diabetes-related symptoms. By analyzing these factors, the proposed system aims to provide a preliminary prediction of whether a person is likely to be diabetic or non-diabetic. The project demonstrates the application of Machine Learning in healthcare and explores how data-driven techniques can support early identification of health risks.

The proposed system uses two classification algorithms, **Random Forest** and **XGBoost**, to develop and compare diabetes prediction models. Before training, the dataset is preprocessed by handling missing or inconsistent values, converting categorical features into numerical form, and dividing the data into training and testing sets. Random Forest uses multiple decision trees to produce reliable classification results, while XGBoost uses boosting to improve prediction by combining decision trees. Both models are trained using the prepared dataset and evaluated using accuracy, precision, recall, and F1-score. Their performance is compared to determine which model provides better prediction results for the dataset. The project shows that Machine Learning can analyze diabetes-related patterns and provide quick preliminary risk predictions. However, the prediction is intended only as a supportive tool and should not replace medical tests, clinical diagnosis, or professional medical advice from healthcare professionals.

2.INTRODUCTION:

Diabetes is a common chronic disease that affects the body's ability to control blood glucose levels and can lead to serious health complications if not detected early. Early prediction of diabetes risk can help individuals seek medical advice and take appropriate preventive measures. With the growth of Machine Learning, healthcare data can be analyzed to identify patterns related to diseases and support early risk prediction. This project, **Early-Stage Diabetes Risk Prediction Using Random Forest and XGBoost Classifiers**, aims to develop a Machine Learning system that predicts whether a person is likely to have diabetes based on symptoms and personal information. The system considers features such as age, gender, frequent urination, excessive thirst, sudden weight loss, weakness, family history, and other diabetes-related symptoms. The dataset is preprocessed by handling missing values and converting categorical data into numerical form before training the models. Two classification algorithms, Random Forest and XGBoost, are used to build and compare prediction models. Their performance is evaluated using accuracy, precision, recall, and F1-score. The project demonstrates how Machine Learning can be applied to healthcare data for early diabetes risk prediction and can serve as a preliminary supportive tool for encouraging timely medical evaluation.

3.SOFTWARE REQUIREMENTS:

- **Operating System:** Windows 10/11 or Linux
- **Programming Language:** Python 3.x
- **Development Environment:** Jupyter Notebook, Google Colab, or Visual Studio Code
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and XGBoost
- **Dataset:** Diabetes symptoms and patient information dataset in CSV format
- **Algorithms:** Random Forest Classifier and XGBoost Classifier
- **Hardware:** Minimum 4 GB RAM, Intel Core i3 or equivalent processor, and 1 GB available storage